


# Contextual Multi Armed Bandits in Bayesian Sequential Decision-Making in Credit Risk Modeling

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# STUDENT'S DECLARATION

I, the undersigned, declare that this research is my original work, except for cited quotations and references that have properly been acknowledged and referenced in accordance with the USIU-Africa requirements. Additionally, this research has not been submitted to any other college, institution, or any other university for examination, award of degree or publication or academic credit.

**Signed ……** **Date …….**

**Name of the Student (ID Number)**

SUPERVISOR APPROVAL

This research proposal has been submitted for examination with my approval as the University Supervisor

**Signed ……** **Date …….**

**Name of the Supervisor (ID Number)**

# DEDICATION

**To me, myself and my ideas.**

To the mind that dared to ask not only *what is likely to happen*, but *what should be done next*.

To the ideas that began as questions, became models, and evolved into decisions, where uncertainty is not something to be feared, but something to be understood, quantified, and learned from.

This work is dedicated to the journey of turning data into knowledge, knowledge into probability, and probability into intelligent decisions. To every uncertainty confronted, every assumption questioned, and every failed approach that became another step toward a better one.

To the idea that decisions should learn from their context.
To the belief that risk can be understood without pretending that uncertainty does not exist.
To the curiosity that led me from predicting outcomes to asking the more difficult question: given what I know now, what should I do next?

To me, for having the courage to explore unfamiliar ideas.
To myself, for continuing when the mathematics became difficult and the answers were uncertain.
And to my ideas, for refusing to remain abstract, pushing instead toward models that learn, adapt, and make decisions under uncertainty.

# ABSTRACT

Traditional credit risk modeling primarily treats lending as a static prediction problem, limiting its ability to adapt to changing borrower behavior, distributional shifts, and the delayed feedback inherent in multi-year loan life cycles. This study proposes a hierarchical Bayesian sequential decision-making framework that reframes credit risk as an adaptive policy optimization problem.

The framework integrates a hierarchical Two-Stage Beta-Beta model for estimating default and recovery probabilities, a Markov Chain Transition Matrix (MCTM) for modeling borrower transitions across credit states, and contextual Thompson Sampling for adaptive lending-policy selection. Compound risk-adjusted rewards incorporate interim repayment behavior to mitigate delayed feedback, while a counterfactual shadow lending engine evaluates alternative loan amounts and interest rates. A Population Stability Index (PSI) Sentinel monitors distributional shifts and acts as a circuit breaker against continued use of unsuitable lending policies.

The framework is empirically evaluated using the Lending Club Loan dataset and deployed on resource-constrained Kubernetes (K3s/Talos) infrastructure using consumer-grade hardware. The study contributes an integrated Bayesian approach to adaptive credit risk modeling that jointly considers borrower risk, policy selection, pricing, and capital allocation. The framework is intended to support more adaptive and potentially financially inclusive lending, although such inclusion outcomes require validation through future real-world deployment.

**Keywords:** Bayesian credit risk; sequential decision-making; Thompson Sampling; multi-armed bandits; MCTM; counterfactual simulation; Population Stability Index; adaptive lending.

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# LIST OF ABBREVIATIONS

**A/B:** A/B Testing. A controlled experimental method that compares two or more strategies to determine which performs better based on observed outcomes.

**API:** Application Programming Interface. A set of rules and protocols that enables different software components or applications to communicate with one another.

**K8s:** Kubernetes. A container orchestration platform used to automate the deployment, scaling, management, and recovery of containerized applications.

**K3s:** K3s Kubernetes. A lightweight Kubernetes distribution designed for resource-constrained environments and edge or distributed computing.

**MAB:** Multi-Armed Bandit. A sequential decision-making framework that balances exploration of uncertain alternatives with exploitation of alternatives known to perform well.

**MCMC:** Markov Chain Monte Carlo. A class of computational methods for drawing samples from complex probability distributions, particularly Bayesian posterior distributions.

**MCTM:** Markov Chain Transition Matrix. A matrix that shows the probability, with each row summing to one, of an entity transitioning from one state to another.

**PSI:** Population Stability Index. A statistical measure used to quantify changes in the distribution of a population or feature between a reference dataset and a subsequent dataset.

**RL:** Reinforcement Learning. A machine learning paradigm in which an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.

**TS:** Thompson Sampling. A Bayesian algorithm for sequential decision-making that selects actions by sampling from their posterior reward distributions.

**MLE:** Maximum Likelihood Estimation. A statistical method that estimates model parameters by maximizing the likelihood of observing the available data.

**MAP:** Maximum A Posteriori. A Bayesian estimation method that identifies parameter values that maximize the posterior probability given the observed data and prior distribution.

**CI:** Confidence Interval. An interval constructed from sample data that provides a range of plausible values for an unknown population parameter under a specified confidence level.

**HDI:** Highest Density Interval. A Bayesian credible interval containing the most probable values of a posterior distribution for a specified probability mass.

**Talos:** Talos Linux. A minimal, immutable Linux operating system designed specifically for running Kubernetes clusters.

**CPU:** Central Processing Unit. The primary processor responsible for executing general-purpose computational instructions.

**GPU:** Graphics Processing Unit. A processor designed for highly parallel computation and commonly used to accelerate machine learning and numerical workloads.

**RAM:** Random Access Memory. Volatile computer memory used to temporarily store data and instructions required by running applications.

**ROI:** Return on Investment. A financial measure used to evaluate the profitability of an investment relative to its cost.

**ROC:** Receiver Operating Characteristic. A performance-analysis curve that illustrates the trade-off between true-positive and false-positive rates for a binary classifier.

**AUC:** Area Under the Curve. A summary measure of classifier discrimination, commonly referring to the area under the ROC curve.

**PSI Sentinel:** Population Stability Index Sentinel. A monitoring mechanism that continuously evaluates distributional divergence and can trigger protective actions when significant population shifts are detected.

**Bayesian Circuit Breaker:** A decision-control mechanism that uses Bayesian uncertainty and distributional-shift information to prevent continued reliance on an incumbent policy when its operating environment changes substantially.

# CHAPTER ONE: INTRODUCTION

## 1.1 Introduction

Credit risk modeling plays a central role in financial decision-making by enabling lenders to estimate the likelihood that a borrower will default and to determine the level of risk associated with extending credit. Traditional credit risk systems, however, are predominantly designed as predictive models that estimate borrower outcomes from historical data. Lending strategies are commonly evaluated using static comparisons, such as A/B testing or historical back-testing, in which alternative strategies are assessed against previously observed outcomes. Although these approaches can provide useful estimates of predictive performance, they are inherently limited in environments where borrower behavior, economic conditions, and portfolio characteristics evolve over time.

The dynamic nature of credit markets introduces several challenges for static modeling approaches. Changes in borrower characteristics and economic conditions can produce distributional shifts between the data used to train a model and the data encountered after deployment. Such shifts may lead to concept drift and deterioration in model performance. Population Stability Index (PSI) provides one mechanism for identifying divergence between the training and deployment distributions; however, detecting distributional change alone does not provide a mechanism for adapting lending policies to the new environment.

A further challenge arises from the delayed nature of credit-risk feedback. The ultimate profitability of a lending decision cannot generally be observed immediately because loans may remain active for several years before their complete repayment or default outcomes become known. Consequently, a lending strategy may continue to generate decisions for a substantial period before sufficient evidence becomes available to determine whether those decisions were financially optimal. This creates a sequential decision-making problem in which actions must be selected under uncertainty while feedback is incomplete and delayed.

This research therefore approaches credit risk from a sequential Bayesian decision-making perspective. Rather than treating credit risk solely as a binary classification problem, the proposed framework models lending as a continuous process of estimating risk, selecting policies, observing borrower behavior, updating uncertainty, and adapting subsequent decisions. The framework integrates a

hierarchical Two-Stage Beta-Beta model, a Markov Chain Transition Matrix (MCTM), and a contextual Thompson Sampling Multi-Armed Bandit.

The hierarchical Bayesian component estimates default probabilities and conditional recovery rates across borrower sub-populations, allowing heterogeneous borrower characteristics to be incorporated into risk estimation while reducing the potential for aggregation bias and Simpson's Paradox. The MCTM complements these terminal outcome estimates by modeling transient borrower transitions through the states of Current, Delinquent, Recovered, and Absorbed Loss. These models provide probabilistic estimates that can be transformed into compound, risk-adjusted reward signals incorporating interim repayment behavior.

The resulting reward signals are supplied to a contextual Thompson Sampling mechanism, which enables the system to balance exploration of potentially profitable lending strategies with exploitation of strategies supported by existing evidence. This allows lending policies to evolve as additional borrower outcomes become available rather than requiring complete loan life cycles before policy adaptation can occur.

The framework additionally incorporates a counterfactual policy simulation mechanism through a shadow lending engine. The engine evaluates alternative loan amounts and interest rates for borrowers independently of their historical lending decisions, enabling the framework to estimate the potential profitability of policies that were not actually observed in the historical dataset. A PSI Sentinel further monitors changes in the feature distribution and functions as a Bayesian circuit breaker, preventing an incumbent strategy from continuing indefinitely when the deployment environment diverges substantially from the training environment.

The framework is evaluated empirically using the Lending Club Loan dataset as a case study. The Lending Club Loan dataset provides a suitable historical environment for investigating borrower risk, loan outcomes, repayment behavior, recovery, and lending profitability. The dataset is not treated as evidence that the proposed system is suitable for immediate real-world autonomous lending. Rather, it provides an empirical setting in which the proposed Bayesian sequential decision-making mechanisms can be developed and evaluated.

The broader intended implication of the research is the development of adaptive credit decision systems capable of identifying risk while potentially expanding appropriate access to credit through more context-sensitive pricing and capital allocation. However, the financial-inclusion implications of the

framework cannot be empirically established using the Lending Club Loan dataset alone. Demonstrating such an effect would require deployment in a real-world lending environment and evaluation across borrowers, institutions, economic conditions, and regulatory contexts. Consequently, financial inclusion is positioned as an intended societal implication of the framework rather than a directly validated outcome of this study.

## 1.2 Problem Statement

Traditional credit risk modeling primarily focuses on predicting whether a borrower will default based on historical borrower and loan characteristics. Although such predictive models can estimate individual or portfolio-level risk, they provide limited mechanisms for continuously adapting the lending policy itself. Lending strategies are often evaluated through static historical comparisons or A/B testing, implicitly treating the underlying borrower population and economic environment as sufficiently stable for the evaluation period.

This assumption is problematic because credit environments are dynamic. Changes in borrower characteristics, economic conditions, and portfolio composition can result in distributional shifts between historical training data and subsequently observed borrowers. When such shifts occur, models and lending strategies calibrated to historical distributions may become progressively less appropriate. Although measures such as the Population Stability Index can identify changes in feature distributions, conventional monitoring systems generally treat drift detection as a diagnostic activity rather than integrating it directly into sequential policy adaptation.

The problem is compounded by the delayed feedback inherent in lending. The true financial outcome of a lending decision may only become observable after months or years of repayment behavior. Consequently, a lender cannot immediately determine whether a particular loan amount, interest rate, or lending strategy was optimal. Static evaluation approaches that rely exclusively on terminal loan outcomes therefore create substantial feedback latency. During this period, sub-optimal policies may continue to be deployed, potentially resulting in sustained portfolio losses.

Existing approaches also face challenges when borrower populations are heterogeneous. Aggregating borrowers into broad risk categories can obscure meaningful differences between sub-populations and may contribute to aggregation bias and Simpson's Paradox. Furthermore, evaluating only the lending decisions historically observed in a dataset prevents direct assessment of alternative policies that were

not selected. A borrower who historically received a particular loan amount at a particular interest rate may have generated a different risk-adjusted return under an alternative exposure or pricing strategy, but this counterfactual outcome is not directly observed.

The resulting research problem is therefore not limited to predicting borrower default. The broader challenge is to develop a credit decision-making framework capable of estimating heterogeneous borrower risk, modeling evolving borrower states, learning from delayed and incomplete feedback, selecting lending policies under uncertainty, evaluating counterfactual policies, and responding to distributional change.

This research addresses this problem by proposing a hierarchical Bayesian sequential decision-making framework that combines Bayesian risk estimation, Markov credit-state modeling, contextual Thompson Sampling, counterfactual policy simulation, and distributional-shift monitoring. The Lending Club Loan dataset is employed as an empirical case study through which the proposed framework can be evaluated.

## 1.3 Objectives

### *1.3.1 General Objective*

To develop and evaluate a hierarchical Bayesian sequential decision-making framework for adaptive credit risk modeling that learns risk-adjusted lending policies under uncertainty, delayed feedback, heterogeneous borrower populations, and distributional shift.

### *1.3.2 Specific Objectives*

1. **To develop a hierarchical Bayesian credit-risk modeling framework** that estimates borrower default, conditional recovery, and credit-state transitions across heterogeneous borrower populations using a Two-Stage Beta-Beta model and MCTM.
2. **To develop an adaptive Bayesian lending-policy optimization mechanism** using compound risk-adjusted rewards and contextual Thompson Sampling to select loan amounts and interest rates under uncertainty and delayed feedback.
3. **To develop a counterfactual and distributional-monitoring framework** incorporating a shadow lending engine and PSI-based monitoring to evaluate alternative lending policies and detect changes in borrower populations.

4. **To evaluate the proposed framework empirically and computationally** using the Lending Club Loan dataset and resource-constrained infrastructure, assessing predictive performance, policy performance, adaptability, profitability, and computational feasibility.

## 1.4 Research Questions

The study seeks to answer the following research questions:

1. How effectively can hierarchical Bayesian models estimate borrower risk and credit-state dynamics across heterogeneous borrower populations?
2. To what extent can contextual Thompson Sampling improve adaptive lending-policy selection using risk-adjusted rewards under delayed feedback?
3. Can counterfactual simulation and distributional monitoring improve lending-policy evaluation and adaptation compared with historically observed lending strategies?
4. How effectively can the proposed framework operate in practice, in terms of predictive accuracy, risk-adjusted profitability, adaptability, and computational feasibility on resource-constrained infrastructure?

## 1.5 Justification

The research is justified by the limitations of treating credit risk exclusively as a static prediction problem. A borrower default probability is useful for estimating risk, but a lending institution ultimately needs to make decisions concerning whether to lend, how much to lend, and at what price. These decisions are sequential and occur under uncertainty, making them fundamentally different from the task of predicting a fixed historical outcome.

A Bayesian framework is particularly appropriate for this problem because lending decisions are made with incomplete information and uncertain future outcomes. Bayesian posterior distributions provide a natural representation of uncertainty and can be updated as additional borrower evidence becomes available. This makes Bayesian inference compatible with sequential decision-making and provides a principled foundation for Thompson Sampling.

The hierarchical structure is justified by the heterogeneity of credit portfolios. Borrowers do not constitute a homogeneous population, and risk characteristics can differ substantially across borrower segments. Estimating parameters hierarchically allows information to be shared across related sub-

populations while preserving segment-level variation. This provides a more appropriate representation of uncertainty than treating all observations as originating from a single homogeneous population.

The inclusion of the MCTM addresses another limitation of conventional default models: borrower credit risk evolves over time. A borrower who is currently current may subsequently become delinquent, recover, or experience absorbed loss. Modeling these intermediate states provides information before terminal loan outcomes become available and therefore creates an opportunity to reduce feedback latency.

The use of Thompson Sampling is justified by the exploration-exploitation trade-off inherent in lending policy selection. A lender must exploit strategies that have demonstrated favorable risk-adjusted returns while retaining the ability to explore alternatives that may perform better for particular borrower contexts. Bayesian Thompson Sampling provides a mechanism for making this trade-off while explicitly incorporating posterior uncertainty.

Counterfactual policy simulation further extends the framework beyond historical policy evaluation. Historical datasets contain outcomes only for decisions that were actually made. A shadow lending engine provides a mechanism for evaluating alternative exposure and pricing strategies, thereby allowing the research to investigate policy opportunities that are hidden by the historical decision process.

The PSI Sentinel is justified by the non-stationary nature of credit environments. A strategy that performs well under one borrower distribution may become inappropriate after a substantial shift in the population. Integrating distributional monitoring with policy control therefore provides an additional safeguard against allowing an incumbent strategy to persist despite changes in the underlying environment.

Finally, evaluating the framework on resource-constrained Kubernetes infrastructure addresses the practical feasibility of deploying sophisticated Bayesian decision systems outside centralized cloud environments. Demonstrating computational autonomy on consumer-grade hardware would provide evidence that advanced sequential credit decision systems do not necessarily require large-scale cloud infrastructure.

The Lending Club Loan dataset provides an appropriate empirical case study for evaluating these mechanisms because it contains historical loan and borrower information together with subsequent loan

outcomes. However, the dataset is used to evaluate the proposed methodology rather than to establish claims about real-world lending effectiveness or financial inclusion. The latter would require prospective deployment, appropriate governance, and empirical evaluation with actual lending decisions and borrower outcomes.

## 1.6 Scope

This research focuses on the development and empirical evaluation of a Bayesian sequential decision-making framework for consumer credit risk and lending-policy optimization.

The empirical component of the study uses the Lending Club Loan dataset as a case study. The analysis focuses on borrower and loan characteristics, loan repayment outcomes, default behavior, recovery, and financial returns that can be derived from the available historical information.

The methodological scope encompasses five principal components:

1. **Hierarchical Bayesian risk estimation**, using a Two-Stage Beta-Beta formulation to estimate default and conditional recovery probabilities across borrower sub-populations.
2. **Credit-state transition modeling**, using a Markov Chain Transition Matrix consisting of Current, Delinquent, Recovered, and Absorbed Loss states.
3. **Sequential policy optimization**, using contextual Thompson Sampling to select lending strategies based on borrower context and posterior uncertainty.
4. **Counterfactual policy evaluation**, using a shadow lending engine to simulate alternative loan amounts and interest-rate decisions.
5. **Distributional-shift monitoring**, using PSI as a sentinel mechanism for detecting changes between the historical training distribution and subsequent borrower populations.

The policy space considered by the framework is centered on lending exposure and pricing, particularly loan amount and interest rate. The framework evaluates these decisions through risk-adjusted profitability rather than treating default prediction accuracy as the sole objective.

The computational scope includes implementation and evaluation on resource-constrained infrastructure using a Kubernetes-based architecture, specifically K3s/Talos and consumer-grade hardware. The objective is to assess computational feasibility rather than to reproduce the infrastructure requirements of a large commercial financial institution.

The study does not constitute a live lending system and does not make actual credit decisions for real borrowers. It does not establish regulatory compliance, operational suitability for financial institutions, or causal effects of lending policies on borrower welfare. Similarly, the research does not empirically establish that the framework improves financial inclusion. Financial inclusion is treated as an intended implication that would require validation through future real-world deployment.

The study is also limited by the observational nature of the Lending Club Loan dataset. Historical data reflect lending decisions that were actually made and therefore cannot fully reveal the outcomes of policies that were never deployed. Counterfactual simulation is consequently used as a modeling mechanism rather than being interpreted as direct observation of counterfactual outcomes.

## 1.7 Expected Contributions

The research is expected to make methodological, computational, and practical contributions to Bayesian credit risk modeling and sequential decision-making.

### *1.7.1 Methodological Contribution*

The primary contribution is a unified framework that connects credit risk estimation with sequential policy optimization. Rather than treating borrower default prediction as the final objective, the framework extends Bayesian modeling into the decision layer, enabling risk estimates to directly influence adaptive lending policies.

The integration of a hierarchical Two-Stage Beta-Beta model with an MCTM is expected to provide complementary representations of credit risk. The Bayesian model captures terminal default and recovery outcomes, while the MCTM captures transient borrower dynamics before loan maturity.

### *1.7.2 Delayed-Feedback Contribution*

The development of compound risk-adjusted reward signals provides a mechanism for incorporating interim repayment behavior into sequential policy learning. This is intended to reduce dependence on terminal loan outcomes and enable policy adaptation before complete loan life cycles have elapsed.

### *1.7.3 Adaptive Policy Contribution*

The integration of contextual Thompson Sampling provides an adaptive mechanism for selecting lending policies under uncertainty. This reframes lending from a static classification problem into a sequential optimization problem involving exploration, exploitation, and posterior learning.

### *1.7.4 Counterfactual Decision Contribution*

The shadow lending engine provides a mechanism for evaluating alternative loan amounts and interest rates that were not historically selected. This enables the framework to investigate whether different pricing and exposure policies could potentially produce improved risk-adjusted profitability.

### *1.7.5 Distributional-Shift Contribution*

The PSI Sentinel introduces a mechanism for incorporating distributional-shift detection directly into policy governance. Rather than treating PSI solely as a monitoring statistic, the proposed framework uses detected distributional divergence to influence whether an incumbent lending strategy should remain active.

### *1.7.6 Computational Contribution*

The implementation on K3s/Talos using consumer-grade hardware is expected to demonstrate the feasibility of running Bayesian sequential decision systems in resource-constrained environments. This contributes to the broader question of whether sophisticated probabilistic decision systems can be computationally autonomous without requiring centralized cloud infrastructure.

### *1.7.7 Practical and Societal Contribution*

The framework is intended to provide a foundation for more adaptive credit underwriting in which pricing and capital allocation can respond to borrower context and changing risk conditions. Such an approach could potentially contribute to more responsible and financially inclusive lending by enabling lenders to distinguish between borrowers who may present different levels of risk rather than relying exclusively on broad static risk categories.

However, this contribution is prospective rather than empirically established within the present study. Demonstrating an actual improvement in financial inclusion would require real-world deployment and

evaluation using measures such as access to credit, approval rates, pricing outcomes, borrower outcomes, portfolio sustainability, and performance across underserved borrower populations.

## 1.8 Chapter Summary

This chapter established the motivation and research foundation for a hierarchical Bayesian framework for sequential decision-making in credit risk modeling. Traditional credit risk approaches were identified as being constrained by static policy evaluation, heterogeneous borrower populations, distributional shift, and the delayed feedback associated with multi-year loan life cycles.

The chapter presented the research problem as a broader sequential decision-making challenge rather than solely a borrower default prediction problem. The proposed framework addresses this challenge by combining hierarchical Bayesian risk estimation, a Markov Chain Transition Matrix, contextual Thompson Sampling, counterfactual policy simulation, and PSI-based distributional monitoring.

The general objective and specific objectives were established to guide the development and empirical evaluation of the framework. Corresponding research questions were formulated around risk estimation, credit-state modeling, delayed feedback, adaptive policy selection, counterfactual evaluation, distributional-shift detection, computational feasibility, and the potential implications for financial inclusion.

The study is scoped around the Lending Club Loan dataset as an empirical case study and does not claim to constitute a real-world autonomous lending system. In particular, financial inclusion is identified as an intended implication rather than an empirically demonstrated outcome. Validation of this implication would require deployment in a real lending environment.

Finally, the expected contributions were identified across methodological, adaptive decision-making, counterfactual evaluation, distributional monitoring, computational, and practical dimensions. Collectively, the research seeks to shift the conceptualization of credit risk from a static prediction task toward an adaptive Bayesian policy optimization problem in which borrower risk, lending strategy, pricing, and capital allocation are continuously learned under uncertainty.

# CHAPTER TWO: LITERATURE REVIEW

## 2.1 Introduction

This chapter reviews the theoretical and empirical literature underpinning sequential decision-making under uncertainty, with particular emphasis on Multi-Armed Bandits (MABs), Bayesian decision-making, Thompson Sampling, contextual bandits, delayed feedback, and their applications to adaptive decision systems. The review establishes the progression from classical MAB problems toward contextual and Bayesian approaches capable of incorporating additional information into sequential decisions.

The chapter begins by examining the general development of Multi-Armed Bandit methods and the exploration-exploitation problem. It then considers Bayesian approaches to bandit optimization, particularly randomized probability matching and Thompson Sampling. Subsequent literature is reviewed on the mathematical properties of Thompson Sampling, including regret, convergence, delayed feedback, and its empirical performance relative to alternative bandit algorithms.

The review further considers the extension of MABs into contextual decision-making, where decisions are conditioned on individual characteristics. This is particularly relevant to credit risk because borrowers possess heterogeneous characteristics that can influence both risk and the profitability of different lending policies. Literature concerning high-dimensional contextual bandits and machine-learning-generated surrogate rewards is therefore examined to establish approaches for decision-making when rewards are delayed or only partially observable.

Finally, the chapter examines the intersection between these sequential decision-making approaches and credit risk modeling. Existing credit-risk literature predominantly focuses on default prediction and risk classification rather than adaptive policy optimization. The chapter consequently identifies the gap between Bayesian credit-risk estimation and sequential lending-policy optimization, providing the basis for the framework proposed in this research.

## 2.2 General Literature Review

### *2.2.1 Multi-Armed Bandits and Sequential Decision-Making*

The Multi-Armed Bandit problem represents a class of sequential decision-making problems in which an agent repeatedly selects among multiple alternatives, or arms, and receives a reward associated with the selected alternative. The fundamental challenge is the exploration-exploitation dilemma. Exploration involves selecting uncertain alternatives to acquire information about their potential performance, while exploitation involves selecting alternatives that are currently believed to provide the greatest reward.

Traditional approaches to the Multi-Armed Bandit problem include deterministic greedy strategies, equal allocation, Upper Confidence Bound (UCB) methods, and the Gittins Index. These approaches differ primarily in how they balance the acquisition of information against the exploitation of currently favorable alternatives.

Scott (2010), in *A Modern Bayesian Look at the Multi-Armed Bandit*, examines randomized probability matching as a Bayesian heuristic for managing Multi-Armed Bandits. The approach selects an arm according to its posterior probability of being optimal. The study compares this strategy with the Gittins Index, UCB algorithms, equal allocation, and deterministic greedy approaches. The principal theoretical motivation is that probability matching provides a simple mechanism for incorporating uncertainty into sequential decisions without requiring the computational complexity of exact optimal policies.

Scott further extends the analysis from the binomial bandit to a fractional factorial bandit in which multiple characteristics influence outcomes. Through logistic or probit regression, a structured model can substantially reduce the number of parameters required compared with treating every possible combination of characteristics as an independent bandit. This provides an important conceptual foundation for contextual bandits, where decisions depend on observable characteristics rather than being selected independently of context.

The relevance of this work to credit decision-making is significant. A lending policy cannot reasonably be evaluated independently of borrower characteristics. Borrower income, loan purpose, credit history, employment characteristics, and other contextual variables may influence the expected reward

associated with a particular lending action. Consequently, the transition from independent MABs to contextual bandits provides a more appropriate framework for heterogeneous credit populations.

### *2.2.2 Bayesian Decision-Making and Thompson Sampling*

Thompson Sampling is a Bayesian approach to the exploration-exploitation problem that selects an action according to samples drawn from the posterior distribution of the expected reward of each action. After observing new outcomes, the posterior distribution is updated, allowing the probability of selecting each action to evolve as evidence accumulates.

Chapelle and Li (2011), in *An Empirical Evaluation of Thompson Sampling*, investigate Thompson Sampling against established approaches including UCB and ε-greedy strategies. The authors argue that the simplicity of Thompson Sampling makes it particularly suitable as a standard baseline for sequential decision-making. The Bayesian formulation also allows the algorithm to naturally incorporate uncertainty into action selection.

The underlying mechanism provides a useful property for adaptive credit decision-making. Rather than permanently selecting the policy with the highest observed historical return, Thompson Sampling continues to assign probability to uncertain policies. This allows potentially superior policies to be explored while increasingly favoring strategies supported by stronger posterior evidence.

The probabilistic nature of the algorithm is particularly relevant when feedback is delayed. In lending, a policy may not receive its complete reward until a loan has matured. A deterministic strategy that immediately commits to the historically best-performing policy may therefore become excessively exploitative based on incomplete information. Thompson Sampling provides a mechanism through which uncertainty can continue to influence policy selection as additional evidence becomes available.

### *2.2.3 Theoretical Properties of Thompson Sampling*

While empirical studies established the effectiveness of Thompson Sampling, Agrawal and Goyal (2011), in *Analysis of Thompson Sampling for the Multi-Armed Bandit Problem*, provide theoretical analysis of its regret properties.

The authors demonstrate that Thompson Sampling achieves logarithmic expected regret for stochastic Multi-Armed Bandit problems. Regret represents the difference between the cumulative reward obtained by the algorithm and the reward that would have been obtained by continuously selecting the

optimal arm. Logarithmic regret implies that the algorithm increasingly concentrates its decisions on the optimal action as additional observations become available.

This theoretical property is important because it provides mathematical support for the empirical observation that Thompson Sampling can approach optimal decision-making over time. The algorithm's posterior updating mechanism allows information acquired through exploration to progressively improve exploitation.

The implications extend to online decision-making environments where decisions must be made before complete outcome information becomes available. This is particularly relevant to credit lending, where the final profitability of a loan may only become observable after an extended period. The ability to update decisions progressively rather than waiting for complete information provides a theoretical basis for applying Thompson Sampling to delayed-feedback environments.

### *2.2.4 Delayed Feedback and Surrogate Rewards*

Delayed feedback represents a fundamental challenge for sequential decision-making. In many real-world systems, the outcome associated with an action cannot be observed immediately. This occurs in credit lending because loan repayment, default, and recovery may occur over extended periods.

Ji et al. (2026), in *Multi-Armed Bandits With Machine Learning-Generated Surrogate Rewards*, investigate the use of machine-learning-generated surrogate rewards when true rewards are delayed or unavailable. Their Machine Learning Assisted Upper Confidence Bound (MLA-UCB) approach uses historical information and side information to generate surrogate rewards that can be incorporated into sequential decision-making.

The study demonstrates that surrogate rewards can provide useful information even when they do not perfectly correspond to the true reward. Under particular statistical conditions, the use of surrogate rewards can reduce regret and preserve asymptotic optimality. Empirical evaluation using language-model selection and video recommendation data further demonstrates improvements over conventional UCB approaches.

The concept of surrogate rewards is particularly relevant to credit risk. Waiting for a loan to mature before determining whether a lending policy was profitable creates substantial feedback latency. Interim repayment behavior, delinquency status, and other borrower-state information may therefore provide useful signals before terminal outcomes are observed.

However, a critical distinction exists between conventional surrogate-reward approaches and the proposed research. Whereas machine-learning-generated surrogate rewards typically approximate an otherwise delayed external reward, the proposed framework derives interim reward signals from explicit Bayesian credit-risk and borrower-state models. The surrogate signal is therefore grounded in probabilistic representations of default, recovery, and credit-state transitions rather than solely in a predictive machine-learning model.

### *2.2.5 Contextual Multi-Armed Bandits*

Contextual Multi-Armed Bandits extend the conventional MAB framework by incorporating information about the environment or individual associated with each decision. Instead of estimating the reward of an arm independently of context, the expected reward is modeled as a function of observed covariates.

Bastani and Bayati (2020), in *Online Decision-Making with High-Dimensional Covariates*, investigate sequential decision-making where individual-level decisions depend on a large number of covariates. The authors introduce a bandit algorithm based on the Least Absolute Shrinkage and Selection Operator (LASSO), allowing the model to identify relevant covariates while performing sequential decision-making.

The study addresses an important challenge in contextual bandits: the covariates observed under a bandit policy are not necessarily independently and identically distributed because the policy itself determines which actions are selected. The authors develop theoretical results establishing convergence of the LASSO estimator under these conditions and demonstrate favorable regret behavior relative to competing contextual-bandit methods.

Their empirical evaluation includes a pharmacological application involving patient characteristics such as demographics, diagnoses, existing medications, and genetic information. The decision concerns the selection of an appropriate treatment, illustrating how contextual information can be used to personalize sequential decisions.

The application provides a strong conceptual parallel to credit lending. Borrowers similarly possess multiple characteristics that may influence the optimal lending decision. Consequently, contextual bandits provide a natural framework for selecting policies based on borrower-specific information rather than applying a single lending strategy uniformly across a portfolio.

### *2.2.6 Bayesian Modeling in Credit Risk*

Conventional credit risk modeling has historically focused on estimating the probability that a borrower will default. Logistic regression, scorecards, tree-based models, and other classification techniques are frequently used to map borrower characteristics to expected credit outcomes.

Bayesian modeling provides an alternative perspective by explicitly representing uncertainty in model parameters and predictions. Rather than producing only point estimates, Bayesian models produce posterior distributions that can be updated as additional evidence becomes available. This property is particularly relevant when credit decisions must be made sequentially and when information arrives over time.

Hierarchical Bayesian models additionally allow parameters to vary across borrower sub-populations while sharing statistical information between groups. Such structures are useful where borrower populations exhibit meaningful heterogeneity but individual groups may contain insufficient observations to support completely independent parameter estimation.

The distinction between predicting risk and making decisions is important in this context. A conventional credit-risk model may estimate that a borrower has a particular probability of default, but the estimate alone does not determine whether the lender should approve the loan, how much capital should be allocated, or what interest rate should be charged. These are policy decisions that depend on the expected reward associated with alternative actions.

The literature therefore establishes a conceptual opportunity for integrating Bayesian credit-risk estimation with sequential decision-making. Such integration would allow posterior risk estimates to directly inform policy selection while preserving uncertainty within the decision process.

## 2.3 Empirical Review

Empirical evidence across the reviewed literature generally supports the effectiveness of Bayesian and contextual bandit approaches for sequential decision-making.

Scott (2010) demonstrates through synthetic experiments that randomized probability matching performs competitively against established approaches across different bandit configurations. The experiments illustrate how posterior distributions evolve as observations accumulate and provide empirical support for using probability matching as a simple Bayesian heuristic.

Chapelle and Li (2011) provide stronger empirical evidence for Thompson Sampling through experiments involving one simulated dataset and two real-world datasets involving display advertising and news recommendation. Thompson Sampling achieved lower regret than the evaluated UCB and ε-greedy approaches across the experiments. The results suggest that the Bayesian sampling mechanism can provide a favorable balance between exploration and exploitation.

Agrawal and Goyal (2011) complement these empirical results with theoretical analysis, establishing logarithmic expected regret for Thompson Sampling in stochastic bandit settings. This provides mathematical support for the empirical observation that the algorithm increasingly identifies and exploits superior actions.

Ji et al. (2026) demonstrate that surrogate rewards can improve sequential decision-making when actual rewards are delayed. Their experiments using language-model selection and video recommendation show that the MLA-UCB approach can outperform conventional UCB. These findings indicate that useful information can be extracted before the true reward becomes fully observable.

Bastani and Bayati (2020) demonstrate the potential of contextual bandits for personalized decision-making in high-dimensional environments. Their synthetic experiments show competitive performance for the LASSO Bandit algorithm, while their pharmacological case study demonstrates that incorporating patient-level covariates can produce better decisions than competing approaches and, in the reported experiment, outperform physicians' decisions.

Taken collectively, these studies demonstrate three important empirical findings. First, Bayesian sampling approaches can achieve competitive or superior regret relative to conventional bandit algorithms. Second, delayed feedback does not necessarily prevent sequential optimization when appropriate probabilistic or surrogate information is available. Third, incorporating contextual information enables policies to be adapted to individual characteristics rather than treating all decision subjects as homogeneous.

Despite these findings, the empirical literature largely evaluates bandit systems in advertising, recommendation, experimentation, and medical decision-making. Credit lending presents a substantially different environment because rewards are influenced by loan pricing, exposure, default, recovery, and borrower-state evolution over potentially multi-year periods.

## 2.4 Research Gaps

The reviewed literature establishes a strong foundation for Bayesian sequential decision-making but reveals several gaps relevant to adaptive credit risk modeling.

### *2.4.1 Separation Between Risk Estimation and Policy Optimization*

Existing credit-risk research predominantly focuses on estimating borrower default probability or classifying borrowers into risk categories. Conversely, Multi-Armed Bandit literature focuses on selecting actions that maximize expected reward. Limited attention has been given to integrating detailed Bayesian credit-risk estimation directly into a sequential lending-policy optimization framework.

The proposed research addresses this gap by connecting borrower-level Bayesian risk estimation to policy selection through risk-adjusted rewards.

### *2.4.2 Limited Integration of Default and Recovery*

Default alone does not completely determine the financial outcome of a loan. Recovery after default materially affects the lender's realized loss. The reviewed bandit literature generally treats rewards as directly observable outcomes rather than decomposing them into default and recovery processes.

This research addresses the gap through a Two-Stage Beta-Beta model that separately estimates default probability and conditional recovery.

### *2.4.3 Limited Modeling of Transient Credit States*

Conventional credit-risk models frequently focus on terminal outcomes such as default or repayment. However, borrowers transition through intermediate states before these outcomes are realized. The reviewed bandit literature does not provide a credit-specific mechanism for incorporating these transient states into sequential rewards.

The proposed MCTM addresses this limitation by modeling transitions through Current, Delinquent, Recovered, and Absorbed Loss states.

### *2.4.4 Delayed Feedback in Credit Lending*

Prior research has established that delayed rewards can be addressed through Bayesian updating and surrogate rewards. However, the specific challenge of multi-year loan life cycles introduces a distinctive form of delayed feedback in which interim borrower behavior can provide information about eventual profitability.

The proposed research extends this literature by constructing compound risk-adjusted rewards from interim credit-state information and Bayesian estimates of eventual default and recovery.

### *2.4.5 Counterfactual Lending Policy Evaluation*

Historical lending datasets primarily contain outcomes associated with decisions that were actually made. Consequently, conventional evaluation cannot directly determine whether an alternative loan amount or interest rate would have produced a better outcome for the same borrower.

The reviewed bandit literature does not adequately address this problem within credit lending. This research introduces a shadow lending engine to simulate alternative lending policies and evaluate their estimated risk-adjusted profitability.

### *2.4.6 Distributional Shift and Policy Governance*

Contextual bandit research generally assumes that the decision environment is sufficiently stable for sequential learning to remain valid. Credit environments, however, can experience substantial changes in borrower characteristics and economic conditions.

The proposed framework addresses this gap through PSI-based distributional monitoring. The PSI Sentinel is incorporated directly into policy governance rather than being treated solely as an offline model-monitoring statistic.

### *2.4.7 Resource-Constrained Deployment*

Much of the literature evaluates sequential decision systems within conventional computational environments. Less attention has been given to the deployment of integrated Bayesian sequential decision systems on resource-constrained infrastructure.

This research therefore extends the implementation dimension by demonstrating the proposed framework using K3s/Talos Kubernetes infrastructure and consumer-grade hardware. The objective is

to establish whether adaptive Bayesian credit decision-making can operate without dependence on centralized cloud infrastructure.

### *2.4.8 Financial Inclusion Requires Real-World Validation*

The reviewed literature provides evidence that personalized and adaptive decision-making can improve outcomes in several domains. However, there is insufficient evidence to conclude that an adaptive Bayesian credit policy will necessarily improve financial inclusion.

The proposed framework may provide mechanisms through which borrowers are evaluated more contextually and lending exposure and pricing are optimized more dynamically. Nevertheless, whether this translates into improved access to credit, fairer pricing, or improved outcomes for underserved populations cannot be established using the Lending Club Loan case study alone.

This therefore represents an important boundary of the current research and a potential direction for future empirical validation through real-world lending deployment.

## 2.5 Chapter Summary

This chapter reviewed the literature underlying the proposed Bayesian sequential credit decision-making framework. The review began with the Multi-Armed Bandit problem and the exploration-exploitation dilemma before examining Bayesian probability matching and Thompson Sampling as approaches for sequential decision-making under uncertainty.

The literature demonstrates that Thompson Sampling is both empirically competitive and theoretically well grounded, with logarithmic regret under stochastic bandit assumptions. Subsequent research demonstrates that Bayesian bandit approaches can operate under delayed feedback and that contextual bandits can incorporate individual-level characteristics into sequential decisions. Research on surrogate rewards further demonstrates that interim information can be used when true rewards are unavailable or delayed.

The empirical review indicates that these approaches have been successfully applied to domains including advertising, recommendation systems, and personalized medical decisions. However, the literature remains fragmented when applied to credit risk. Existing approaches generally separate risk estimation from policy optimization and provide limited treatment of recovery, transient borrower states, counterfactual lending policies, and distributional shift.

The identified gaps motivate the proposed research framework, which integrates hierarchical Bayesian default and recovery modeling, MCTM-based borrower-state transitions, contextual Thompson Sampling, counterfactual shadow lending, and PSI-based monitoring. The framework therefore extends the existing literature from static credit-risk prediction toward adaptive sequential optimization of lending policies.

The next chapter presents the methodology used to develop, implement, and evaluate the proposed framework.

# CHAPTER THREE: METHODOLOGY

## 3.1 Introduction

In credit lending, financial institutions deal with the question of “Given the risk profile of this population, what lending strategy produces the greatest risk-adjusted revenue?” To handle this question I propose a three pronged mathematical modeling approach with the end product being an open source research prototype deployed as a containerized workload on a Kubernetes cluster.

Firstly, the risk profile of a population is modeled using a Two-Stage Beta-Beta framework. Moreover, the exact behavior of each population is understood as a Markov Chain Transition Matrix (MCTM) with absorbing states. Finally, to determine the lending strategy with the greatest risk-adjusted revenue a Gaussian Multi-Armed Bandit is leveraged.

To address Simpson's Paradox and aggregation bias, the Bayesian models are formulated hierarchically, allowing borrower outcomes to be estimated across different population segments and lending policies. A lending strategy that appears profitable when evaluated over the entire portfolio may perform poorly when evaluated within specific borrower groups. Therefore, the framework models outcomes conditional on both borrower characteristics and the applied lending action.

## 3.2 Design Methodology

The research adopts a design science methodology to develop and evaluate an adaptive Bayesian credit decision-making prototype. The methodology is centered on the design, implementation, and empirical evaluation of a computational artifact that addresses the limitations of static credit-risk modeling. Rather than developing an isolated predictive model, the research integrates multiple statistical and computational components into a unified sequential decision-making system.

The proposed design follows a layered architecture consisting of risk estimation, borrower-state modeling, policy optimization, distributional monitoring, and counterfactual simulation. These components operate together to transform historical borrower and loan information into adaptive lending decisions.

At the first layer, borrower and loan characteristics are used to establish the risk profile of the lending population. A hierarchical Two-Stage Beta-Beta model estimates the probability of default and the

conditional recovery rate for different borrower groups and lending policies. The hierarchical formulation allows information to be shared across groups while retaining group-specific behavior, reducing the instability that can arise when individual populations contain limited observations. It also allows the analysis to explicitly condition risk estimates on both borrower characteristics and the lending policy applied.

The second layer models the temporal behavior of borrowers using a Markov Chain Transition Matrix. Borrowers are represented as occupying one of four credit states: Current, Delinquent, Recovered, or Absorbed Loss. The transition structure provides estimates of how borrower behavior evolves throughout the loan life cycle. Recovered and Absorbed Loss represent terminal states, while Current and Delinquent represent transitional states. This component therefore provides information about borrower behavior before the final loan outcome is observed.

The third layer transforms the probabilistic risk estimates into risk-adjusted reward distributions. Instead of treating default as the sole objective, the reward function considers loan exposure, interest revenue, expected losses, and recovery. Interim borrower-state information obtained from the MCTM can therefore contribute to the reward before the loan reaches maturity. This design reduces the dependence of the decision-making system on delayed terminal outcomes.

The fourth layer performs sequential lending-policy optimization using a Multi-Armed Bandit. Each arm represents a candidate lending policy defined by a combination of loan amount and interest rate. At each decision epoch, Thompson Sampling draws a value from the posterior reward distribution associated with each policy and selects the policy with the highest sampled value. After additional borrower outcomes become available, the posterior distribution of the selected policy is updated. This creates a continuous feedback loop in which the system progressively learns which lending policies produce the greatest risk-adjusted returns for different borrower contexts.

The fifth layer introduces a counterfactual shadow lending engine. Historical lending data only contains outcomes for policies that were actually implemented. Consequently, the observed dataset cannot directly establish what would have happened if a borrower had received a different loan amount or interest rate. The shadow engine addresses this limitation by using posterior predictive distributions from the Bayesian models to simulate alternative lending policies. These simulations allow the expected risk-adjusted profitability of unobserved policies to be compared with the historical lending policy.

The sixth layer provides distributional monitoring and policy governance through a Population Stability Index Sentinel. The Sentinel continuously compares the distribution of incoming borrower characteristics against the reference training distribution. When substantial divergence is detected, the system can trigger a Bayesian circuit-breaker mechanism that prevents an incumbent policy from continuing without review. This provides protection against concept drift and changes in the underlying borrower population. The resulting human-in-the-loop process can also trigger reassessment of the population segmentation used by the hierarchical models.

The complete methodology therefore follows a closed-loop architecture:

**Borrower Context → Bayesian Risk Estimation → Credit-State Prediction → Risk-Adjusted Reward → Policy Selection → Observed Outcomes → Posterior Update → Policy Adaptation**

The PSI Sentinel operates across this loop as a monitoring and governance mechanism, while the shadow lending engine operates alongside the historical policy to evaluate counterfactual decisions.

### *3.2.1 Methodological Design Principles*

The proposed system is guided by several design principles.

First, uncertainty is explicitly modeled. Bayesian posterior distributions are used rather than relying exclusively on point estimates. This allows uncertainty in default, recovery, and policy profitability to influence subsequent decisions.

Second, borrower heterogeneity is preserved. The hierarchical structure allows the system to model differences between borrower populations and lending policies while sharing information across groups where appropriate. This is intended to reduce aggregation bias and the potential effects of Simpson's Paradox.

Third, decisions are treated as sequential. Lending policies are not assumed to be fixed throughout the lifetime of the portfolio. The system updates its beliefs as new borrower information becomes available and consequently adapts subsequent policy selection.

Fourth, delayed outcomes are supplemented with interim information. The MCTM provides information about borrower-state transitions before terminal outcomes are available, allowing the decision process to incorporate partial evidence rather than waiting for complete loan maturity.

Fifth, policy evaluation extends beyond observed historical decisions. The shadow lending engine provides a counterfactual simulation environment in which alternative loan amounts and interest rates can be evaluated probabilistically.

Sixth, adaptation is subject to distributional monitoring. The PSI Sentinel provides a control mechanism that identifies when the borrower population has diverged sufficiently from the historical training environment to warrant policy reassessment.

Finally, the statistical framework is designed for computational deployment. The resulting system is implemented as a modular, containerized software prototype so that individual modeling and decision components can be independently deployed, monitored, and recovered within a Kubernetes environment.

### *3.2.2 Research Artifact*

The primary artifact produced by the research is an open-source prototype implementing the proposed Bayesian sequential lending framework. The prototype integrates the statistical models, policy optimization mechanism, counterfactual simulator, monitoring mechanisms, and supporting infrastructure into a reproducible computational system.

The artifact is designed as a modular collection of containerized services rather than a single monolithic application. This permits individual components to be independently tested, scaled, replaced, or recovered following infrastructure failures. The deployment architecture described in subsequent sections uses Kubernetes orchestration to provide resilience and operational management.

The design methodology therefore evaluates the research contribution at two levels. At the statistical level, the study evaluates whether the proposed models can accurately estimate borrower risk and identify profitable lending policies. At the systems level, it evaluates whether these models can operate as an adaptive decision-making system under delayed feedback, distributional shift, and resource-constrained infrastructure.

This design provides the foundation for the mathematical formulation, experimental procedures, and evaluation methods presented in the subsequent sections.

## 3.3 Research Design

Let $X_i$ denote the characteristics of borrower $i$, and let $A_i$ denote the lending policy applied, consisting of the loan amount and interest rate:

$$A_i = (L_i, r_i)$$

The first component of the Two-Stage Beta-Beta framework models default behavior using a hierarchical Beta-Bernoulli formulation. For borrower group $g$ under lending policy $a$, the default probability is defined as:

$$p_{g,a} = P(D = 1|X_g, A_a)$$

where $D = 1$ represents a default event. The global portfolio-level default behavior provides the prior distribution from which group and policy-specific parameters are estimated

$$p_0 \sim Beta(\alpha_0, \beta_0)$$

$$p_g \sim Beta(\alpha_g, \beta_g)$$

$$p_{g,a} \sim Beta(\alpha_{g,a}, \beta_{g,a})$$

This hierarchical structure allows borrower groups and lending policies with limited observations to shrink towards the broader portfolio distribution rather than producing unstable estimates from small samples. For a given borrower group $g$ and lending policy $a$, let $d_{g,a}$ represent observed defaults and $n_{g,a}$ observed non-defaults, the posterior distribution is updated as

$$p_{g,a}|D \sim Beta(\alpha_{g,a} + d_{g,a}, \beta_{g,a} + n_{g,a})$$

with posterior mean

$$E[p_{g,a}] = \frac{\alpha_{g,a} + d_{g,a}}{\alpha_{g,a} + \beta_{g,a} + d_{g,a} + n_{g,a}}$$

The second stage models recovery behavior conditional on default. It answers the question: "Given that a borrower defaults, what proportion of the outstanding exposure is expected to be recovered?" The recovery rate is modeled as

$$R_{g,a}|D = 1 \sim Beta(\alpha^R_{g,a}, \beta^R_{g,a})$$

where the Beta parameters are estimated using the Method of Moments approximation. Given the observed recovery rate mean $\mu$, variance $\sigma^2$, and number of observations $N$, the parameters are estimated as

$$\alpha_{g,a}^{R} = \mu \left( \frac{\mu(1-\mu)}{\sigma^2} - 1 \right)$$

$$\beta_{g,a}^{R} = (1-\mu) \left( \frac{\mu(1-\mu)}{\sigma^2} - 1 \right)$$

The resulting posterior predictive distributions provide estimates of default likelihood and expected recovery under alternative lending policies. These estimates form the basis for calculating risk-adjusted reward distributions, which are subsequently provided to the Thompson Sampling Multi-Armed Bandit for sequential lending policy optimization.

Working in tandem with this Two-Stage Beta-Beta framework is the MCTM with the states: Current, Delinquent, Recovered, and Absorbed Loss. Recovered and Absorbed Loss are absorbing states. The MCTM is a predictive framework offering probabilistic predictions on the expected behavior of a borrower given previous data with the state space

$$S = \{C, D, R, L\}$$

and transition matrix

$$P = \begin{bmatrix} p_{CC} & p_{CD} & 0 & 0 \\ p_{DC} & p_{DD} & p_{DR} & p_{DL} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Since Recovered and Absorbed Loss are absorbing states

$$P_{RR} = 1, \qquad P_{LL} = 1$$

the borrower must eventually reach one of the absorbing states.

$$\lim_{k \to \infty} \pi_k = [0, 0, \pi_R^{\infty}, \pi_L^{\infty}]$$

At any time $t$ that is not the maturity time (a transitional state)

$$\pi_t = [\pi_C(t), \pi_D(t), \pi_R(t), \pi_L(t)]$$

the probability of default is

$$P(\text{Default}) = \pi_D(t) + \pi_L(t)$$

since there is still uncertainty about whether the delinquent borrower will repay the loan. However, as loan tends towards maturity the delinquents collapse to either recovery or a loss. Thus the MCTM and Two-Stage Beta-Beta converge and tell the same narrative.

$$\pi_D(t) \to 0$$

$$\lim_{t\to\infty} \pi_R(t) = \mathbb{E}[R]$$

$$\lim_{t\to\infty} \pi_L(t) = \mathbb{E}[p_d]$$

$$\lim_{t\to\infty} |\pi_R(t) - \mathbb{E}[R]| = 0$$

$$\lim_{t\to\infty} |\pi_L(t) - \mathbb{E}[p_d]| = 0$$

The MCTM and Beta-MoM serve different purposes in this framework. The MCTM predicts the probability of recovery or loss depending on delinquency at a give time $t$ which changes dynamically.

$$P(R \mid D, t)$$

$$P(L \mid D, t)$$

The Beta-MoM on the other hand approximates the long run quantities $p_d$ and $R$ without modeling intermediate borrower states. The MCTM informs decision making in transitional states and is invaluable for reporting and forecasts while the Two-Stage Beta-Beta is used as the sampling distribution for the automated decisions of the exact policy to apply given the risk appetite of the credit organization and the risk profile of the borrower.

The lending optimization problem is formulated as a stochastic Multi-Armed Bandit (MAB), where each arm represents a candidate lending policy. A lending policy is defined as a combination of loan exposure and pricing parameters

$$A_k = (L_k, r_k)$$

where $L_k$ represents the loan amount and $r_k$ represents the interest rate.

Let

$$\mathcal{A} = \{A_1, A_2, \ldots, A_K\}$$

denote the set of candidate lending policies, where each policy corresponds to one arm of the Multi-Armed Bandit. Each arm has an unknown expected risk-adjusted reward

$$\theta_k$$

representing the long-term profitability of applying policy $A_k$. The posterior belief over this reward is modeled as

$$\theta_k \sim \mathcal{N}(\mu_k, \sigma_k^2)$$

where $\mu_k$ represents the posterior mean expected risk-adjusted revenue and $\sigma_k^2$ represents the uncertainty associated with the policy. At each decision epoch $t$, Thompson Sampling draws a sample from each posterior distribution

$$\theta_k \sim \mathcal{N}(\mu_k, \sigma_k^2)$$

and selects the lending strategy

$$A_t = \arg\max_k \tilde{\theta}_k$$

The observed reward is defined as a compound risk-adjusted revenue signal derived from Bayesian estimates of repayment behavior

$$R_t = \text{Revenue}_t - \text{Expected Loss}_t - \text{Funding Cost}_t$$

The expected loss component incorporates default probability and recovery uncertainty

$$E[L_t] = L_t P(D|X, A_t)(1 - E[R|X, A_t])$$

After observing borrower repayment information, the posterior distribution of the selected lending policy is updated. Let $n_k$ denote the number of repayment observations for strategy $A_k$ the posterior mean is updated incrementally as

$$\mu_k^{(t+1)} = \mu_k^{(t)} + \frac{r_t - \mu_k^{(t)}}{n_k + 1}$$

where

$$n_k \leftarrow n_k + 1$$

The variance is updated using the online variance estimator

$$\sigma_k^{2(t+1)} = \frac{(n_k - 1)\sigma_k^{2(t)} + (r_t - \mu_k^{(t)})(r_t - \mu_k^{(t+1)})}{n_k}$$

which continually refines the uncertainty associated with each lending strategy as additional borrower outcomes are observed. As the number of observations increases

$$\lim_{n_k \to \infty} \mu_k^{(t)} = \mathbb{E}[r_k]$$

$$\lim_{n_k \to \infty} \sigma_k^2 = \mathrm{Var}(r_k)$$

Thus, the Thompson Sampling agent progressively concentrates probability mass on lending policies with the highest expected risk-adjusted profitability while maintaining exploration of uncertain policies.

In addition, to ensure statistical integrity, a Population Stability Index (PSI) Sentinel will monitor live data streams to detect feature-space deviations from the training distribution. This acts as a Bayesian circuit breaker, preventing a "Champion" model from persisting when the underlying economic ground truth has shifted. This automatically prompts the human in the loop to reevaluate the grouping criteria used to segment the population. Changes in economic ground truth might necessitate further fragmentation of established groups to guard against aggregation bias skewing results and causing significant losses to the credit institution. Similarly, previously dissimilar groups might converge to having similar behaviors leading to operational inefficiencies in modeling separate frameworks for them.

## 3.4 Data Acquisition

To validate my prototype I will use the Lending Club data publicly accessible on Kaggle. To access this data I will explicitly request consent from Kaggle in order to use the data and abide by their strict policies on the use of that data utilizing it only for academic purposes and to advance research into Bayesian Contextual Bandits.

## 3.5 Experimental Procedures

The proposed framework will be implemented using Python, following clean object-oriented software engineering principles and incorporating comprehensive exception handling to manage numerical instability and mathematical boundary conditions inherent in Bayesian inference. The system

architecture will adopt a microservices-based design, with individual components containerized using Docker and deployed on a local Kubernetes cluster. Event-driven state management will be implemented through Apache Kafka using an append-only architecture, enabling asynchronous processing and decoupling model computation from long-term data persistence.

The experimental environment will utilize publicly available historical Lending Club Loan datasets containing borrower characteristics, loan attributes, repayment behavior, and loan outcomes. Model development will leverage Python scientific computing libraries including NumPy, SciPy, pandas, and DuckDB for numerical computation, data processing, and analytical workflows. Workflow orchestration will be managed using Temporal's Python SDK to coordinate distributed model execution and ensure reliable processing.

The deployment infrastructure will consist of a resource-constrained bare-metal Kubernetes cluster managed using Talos Linux and the Sidero Omni control plane. The target environment consists of distributed consumer-grade hardware providing approximately 12-16 CPU cores and 32-44GB of aggregate RAM across worker nodes. Kubernetes management will utilize Helm and Kubernetes CLI tooling for service deployment and life cycle management. System performance and resource utilization will be monitored through a Prometheus and Grafana observability stack, enabling evaluation of computational efficiency and operational reliability.

## 3.6 Data Analysis Methods

My prototype will work by simulating multiple years of data and common challenges experienced in the African market like power cuts, limited financial resources, and common technical complications like failing hardware.

To accomplish this my simulation will intentionally have a ‘Chaos Monkey’ break things. To simulate technical failure some nodes in the cluster will be turned off for minutes or hours at a time. Workloads will be canceled as they are running and the typical black out will be embraced to validate the capability of my prototype to real world conditions.

Beyond evaluating lending strategies on observed outcomes, this research explores the counterfactual performance of alternative lending policies through Bayesian simulation. The underlying premise is that creditworthiness is not solely an intrinsic property of the borrower but is also influenced by the lending policy applied, including loan exposure and pricing decisions. In alignment with Sustainable

Development Goals (SDGs) 1 (No Poverty), 8 (Decent Work and Economic Growth), 10 (Reduced Inequalities), and 17 (Partnerships for the Goals), the framework investigates whether broader financial inclusion can be achieved while maintaining long-term portfolio profitability.

The historical dataset serves as the behavioral baseline, containing borrowers who received credit under an existing lending policy, including borrower characteristics, loan amount, interest rate, repayment behavior, and recovery outcomes. Since the dataset only contains observed lending decisions, the true outcomes under alternative lending policies are unavailable. Therefore, the historical policy is treated as the reference policy, while the proposed framework operates as a shadow lending engine that estimates the expected outcomes of alternative policies through Bayesian posterior predictive modeling.

The shadow lending engine independently determines candidate loan amounts and interest rates for borrowers and estimates the associated risk-adjusted profitability. Using the hierarchical Two-Stage Beta-Beta framework and the Markov Chain Transition Matrix, the simulator models default probability, recovery behavior, and borrower transitions through the credit life cycle. These probabilistic estimates are transformed into compound reward distributions that allow alternative lending policies to be evaluated before complete loan maturity.

Rather than predicting default alone, the framework evaluates whether different combinations of loan exposure and pricing could improve capital allocation by transforming historically unprofitable lending decisions into profitable outcomes, or by identifying opportunities where successful borrowers could have supported increased credit exposure while maintaining acceptable risk-adjusted returns.

The objective is therefore not to maximize approvals or minimize defaults independently, but to learn adaptive lending policies that jointly optimize loan amount, interest pricing, expected recovery, financial inclusion, and long-term portfolio profitability. By framing lending as a sequential Bayesian decision-making problem, the proposed simulator enables the evaluation of lending strategies that were never deployed historically while explicitly accounting for uncertainty in counterfactual outcomes.

## 3.7 Validation and Evaluation

The proposed framework is evaluated from both predictive and decision-making perspectives. The predictive performance of the hierarchical Bayesian models is assessed through calibration and accuracy metrics. The Beta-Bernoulli default model is evaluated using metrics such as the Brier Score,

calibration error, and receiver operating characteristic area under the curve (ROC-AUC) to determine whether estimated default probabilities accurately represent observed borrower outcomes. The Beta-MoM recovery model is evaluated using mean absolute error (MAE) and root mean squared error (RMSE) to measure the accuracy of expected recovery estimates. The Markov Chain Transition Matrix is evaluated by comparing predicted borrower state transitions against observed transitions using transition likelihood and prediction error.

Beyond predictive accuracy, the primary evaluation focuses on decision quality. The historical lending policy is compared against the Bayesian shadow policy using counterfactual risk-adjusted profitability, measuring whether alternative loan amounts and interest rates produce improved expected returns while maintaining acceptable risk exposure. The Thompson Sampling Multi-Armed Bandit is evaluated through cumulative regret, reward convergence, and policy selection performance to determine whether the algorithm successfully identifies higher-performing lending strategies over time. Additionally, financial inclusion outcomes are evaluated by measuring whether the framework can expand responsible credit access while preserving portfolio profitability.

## 3.8 Ethical Considerations

Since credit decisions directly influence financial access and economic opportunity, ethical considerations are central to the proposed framework. Historical lending data may contain existing biases caused by unequal access to financial services, meaning that models trained solely on past decisions may unintentionally reproduce discriminatory patterns. The hierarchical Bayesian structure helps mitigate this risk by preventing small borrower populations from being evaluated using unstable estimates and by explicitly modeling differences across borrower groups. Fairness analysis is performed to ensure that improved profitability does not come at the cost of systematically disadvantaging specific populations.

Furthermore, the optimization objective is designed around responsible lending rather than profit maximization alone. The reward function incorporates risk-adjusted profitability while considering financial inclusion objectives, ensuring that the system does not simply minimize lending risk by excluding higher-risk borrowers. Model explainability is also considered essential, as borrowers and financial institutions require understandable reasoning behind recommended lending policies and pricing decisions.

The proposed framework operates on sensitive financial information; therefore, privacy preservation is a critical consideration. The dataset will undergo appropriate anonymization and removal of personally identifiable information before model development. Data minimization principles will be applied by retaining only variables necessary for estimating borrower behavior and optimizing lending policies. During deployment, secure storage, access control mechanisms, encryption, and audit logging will be enforced to prevent unauthorized access to financial information.

Future implementations may incorporate privacy-preserving techniques such as differential privacy or federated learning to enable collaborative model development across financial institutions without exposing individual borrower records. These approaches would allow the framework to benefit from broader lending data while maintaining borrower confidentiality.

## 3.9 Summary

This research proposes a hierarchical Bayesian decision framework for adaptive credit policy optimization. Historical lending data is first used to estimate borrower behavior through a Two-Stage Beta-Beta model and a Markov Chain Transition Matrix. The Bayesian framework models default probability, recovery behavior, and borrower life cycle transitions conditional on borrower characteristics and lending policies, producing posterior predictive distributions of expected outcomes under alternative lending decisions.

These predictive distributions are transformed into risk-adjusted compound rewards that serve as the input for a Thompson Sampling Multi-Armed Bandit. During the online decision phase, the bandit treats different combinations of loan amount and interest rate as competing lending strategies and sequentially selects policies while updating its beliefs from observed borrower repayment signals. A Population Stability Index (PSI) Sentinel monitors distributional shifts in borrower characteristics to identify potential model drift and prevent outdated lending strategies from remaining deployed.

The complete framework is evaluated through predictive accuracy, counterfactual policy comparison, regret analysis, profitability improvement, and financial inclusion measures. The research demonstrates how hierarchical Bayesian modeling and adaptive decision algorithms can enable more responsive, inclusive, and statistically grounded credit allocation systems.

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# APPENDICES

## Appendix A: Introduction Letter

RE: INTRODUCTION TO THE PROPOSED RESEARCH STUDY

Dear Sir/Madam,

I am a student at the United States International University-Africa undertaking a research study titled **"Contextual Multi-Armed Bandits in Bayesian Sequential Decision-Making in Credit Risk Modeling."**

The proposed study seeks to develop a hierarchical Bayesian framework for adaptive credit policy optimization. The framework will combine a Two-Stage Beta-Beta model for estimating default and recovery, a Markov Chain Transition Matrix (MCTM) for modeling borrower credit-state transitions, and a Contextual Thompson Sampling Multi-Armed Bandit for selecting lending policies based on borrower characteristics and risk-adjusted expected returns.

The study will use the Lending Club Loan dataset as a secondary-data case study. The research will investigate whether sequential Bayesian decision-making can provide a more adaptive approach to credit policy optimization than conventional static credit-risk modeling. Counterfactual simulation will also be explored to evaluate alternative loan amounts and interest-rate policies that were not historically observed.

The proposed research will be conducted for **academic purposes** and will not involve making real-world lending decisions or directly interacting with borrowers. The study will rely on historical data and computational simulations, with appropriate consideration given to data privacy, ethical use of financial information, and responsible lending.

The resulting research prototype is intended to demonstrate the feasibility of integrating Bayesian statistical modeling with adaptive decision-making and computational infrastructure. Any conclusions regarding financial inclusion or real-world lending effectiveness will be treated as potential implications requiring further validation through actual deployment.

I kindly request that this letter serve as an introduction to the proposed research and its objectives. The study will be conducted in accordance with the requirements and ethical guidelines of the institution.

Yours faithfully,

**Misati Randy Nyambane**
Student, Data Science and Analytics
United States International University-Africa

**Supervisor:** Professor Gabriel Okello
**Department:** Computing and Informatics
**Date:** 24th August 2026

# Appendix B: Debrief Letter

RE: DEBRIEFING INFORMATION FOR THE PROPOSED RESEARCH STUDY

Dear Sir/Madam,

This letter provides additional information regarding the proposed research study titled **"Contextual Multi-Armed Bandits in Bayesian Sequential Decision-Making in Credit Risk Modeling."**

The study proposes the development of a hierarchical Bayesian framework for adaptive credit policy optimization. The framework will combine Bayesian estimation of default and recovery, borrower credit-state transitions using a Markov Chain Transition Matrix (MCTM), and contextual Thompson Sampling for sequential lending-policy selection.

The study will use the Lending Club Loan dataset as a secondary-data case study. No borrowers will be directly recruited, interviewed, or asked to provide additional information. The research will therefore focus on statistical modeling, simulation, and computational experimentation using historical observations.

The proposed framework will be evaluated through predictive modeling, counterfactual policy simulation, sequential decision-making experiments, and computational evaluation. The study will investigate whether the framework can identify lending policies that provide improved risk-adjusted outcomes while adapting to changes in borrower characteristics and observed repayment behavior.

The research prototype is intended **solely for academic and experimental purposes**. It will not be used to make actual lending decisions, approve or reject real borrowers, determine real interest rates, or provide financial advice. Any potential implications for financial inclusion, responsible lending, or real-world credit operations will be considered as areas for future investigation rather than outcomes directly established by this study.

All data will be handled in accordance with applicable institutional research requirements and relevant data-use restrictions. The study will apply data minimization and appropriate security measures when processing the secondary dataset.

The findings of the study will contribute to research on Bayesian sequential decision-making, contextual Multi-Armed Bandits, and adaptive credit risk modelling, while also identifying limitations and areas requiring further validation before any real-world deployment.

Yours faithfully,

**Misati Randy Nyambane**
Student, Data Science and Analytics
United States International University-Africa

**Supervisor:** Professor Gabriel Okello
**Department:** Computing and Informatics
**Date:** 24th August 2026

# Appendix C: Work Plan

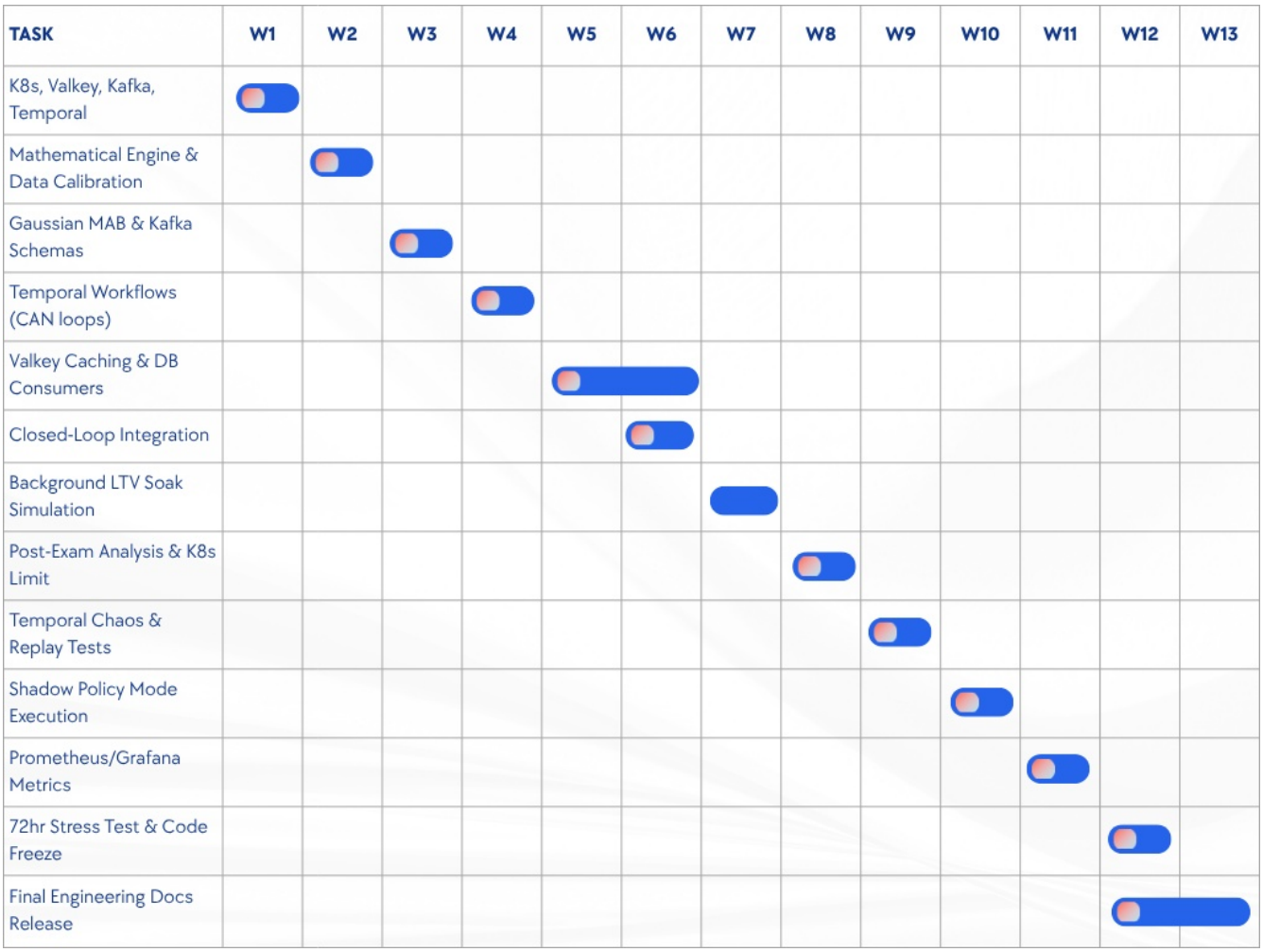



*Figure 1: Research Gantt Chart*

## Appendix D: Data Use and Ethical Compliance

The study will use the **Lending Club Loan dataset as secondary data** for academic research. No direct interaction with borrowers will take place, and the research will not make real-world lending decisions. Data will be used only for the purposes defined in the proposal, with appropriate measures applied to protect privacy, maintain data security, and comply with applicable data-use and institutional ethical requirements.

*Table 1: Data Use and Ethical Compliance Issues*

| Area | Proposed Measure |
|---|---|
| **Data Source** | Publicly available Lending Club Loan dataset |
| **Purpose** | Academic research and development of a Bayesian credit-risk prototype |
| **Participant Interaction** | No direct recruitment or interaction with borrowers |
| **Data Privacy** | Personally identifiable information will not be used where unnecessary |
| **Data Minimization** | Only variables necessary for modeling and evaluation will be retained |
| **Data Security** | Data will be stored in a controlled and access-restricted research environment |
| **Data Sharing** | Data will not be redistributed where restricted by the applicable data-use terms |
| **Ethical Use** | The dataset will not be used for actual lending or automated financial decisions |
| **Bias and Fairness** | Potential bias in historical lending data will be assessed during analysis |
| **Human Oversight** | Any future real-world application would require appropriate human review |
| **Research Scope** | Findings will be presented as experimental evidence and not as proof of real-world lending effectiveness |

## Appendix E: Proposed Research Budget

The proposed research will primarily utilize **open-source software, publicly available secondary data, and repurposed computing hardware**. Older school computers will be requested for use as Kubernetes worker nodes, while Sidero Omni will be hosted on a cloud VPS for cluster management. The budget therefore focuses on infrastructure, networking, power, and contingency costs rather than commercial software or new high-performance computing equipment.

*Table 2: Proposed Research Budget*

| Item | Description | Estimated Cost (KES) |
|---|---|---|
| **Repurposed computing hardware** | Older school computers to be requested and configured as Kubernetes worker nodes | 0 |
| **Network equipment** | Ethernet cables, switches, adapters, and related networking requirements | 5,000 |
| **Cloud VPS** | VPS for Sidero Omni cluster management/control-plane services | 6,000 |
| **Storage** | Additional SSD/HDD storage where required for datasets, logs, and experimental artifacts | 8,000 |
| **Electricity** | Power consumption associated with running the local Kubernetes cluster during experiments | 10,000 |
| **Internet connectivity** | Internet access for package installation, container images, cluster management, and research activities | 6,000 |
| **Software licenses** | Python, Kubernetes, Talos Linux, Helm, Temporal, Prometheus, Grafana, and other open-source tooling | 0 |
| **Data acquisition** | Lending Club Loan dataset | 0 |
| **Contingency** | Unexpected hardware, networking, or | 5,000 |

| Item | Description | Estimated Cost (KES) |
|---|---|---|
| | infrastructure expenses | |
| **Total Estimated Budget** | | **40,000** |

### *Budget Justification*

The proposed budget is deliberately designed around **low-cost and resource-constrained infrastructure**. Rather than purchasing dedicated servers, older school computers will be requested and repurposed as Kubernetes worker nodes. This supports the research objective of evaluating whether the proposed Bayesian framework can operate on modest computing infrastructure.

The majority of the software stack is open source, resulting in **no anticipated software licensing costs**. A small cloud VPS will be used to host the Sidero Omni management component, while the computational workloads will primarily execute on the local bare-metal Kubernetes cluster.

The final costs may vary depending on the availability and condition of the donated or repurposed computers, the selected VPS configuration, and the networking requirements identified during implementation.

## Appendix F: Data Collection and Variable Specification

### *F.1 Data Source*

The study will use the publicly available Lending Club Loan dataset as secondary data. The dataset contains borrower characteristics, credit-history information, loan characteristics, repayment behavior, and final loan outcomes. No primary data will be collected directly from borrowers.

The variables retained for the study will be organized according to their **availability within the lending life cycl**e. This temporal separation is important for preventing data leakage and ensuring that the contextual bandit makes decisions using only information that would have been available at the time of lending.

*Table 3: Retained Variables and Their Availability in the Credit Decision Process*

| Variable | Availability Stage | Primary Use |
|---|---|---|
| annual_inc | Pre-decision | Borrower context / risk estimation |
| annual_inc_joint | Pre-decision | Joint-borrower risk estimation |
| application_type | Pre-decision | Borrower context |
| dti | Pre-decision | Risk estimation |
| dti_joint | Pre-decision | Joint-borrower risk estimation |
| emp_length | Pre-decision | Borrower context |
| fico_range_low | Pre-decision | Credit-risk estimation |
| fico_range_high | Pre-decision | Credit-risk estimation |
| home_ownership | Pre-decision | Borrower context |
| purpose | Pre-decision | Loan context |
| verification_status | Pre-decision | Information-quality assessment |
| verified_status_joint | Pre-decision | Joint-borrower information quality |
| delinq_2yrs | Pre-decision | Historical credit risk |
| delinq_amnt | Pre-decision | Credit risk |
| inq_last_6mths | Pre-decision | Credit-seeking behavior |

| Variable | Availability Stage | Primary Use |
|---|---|---|
| inq_last_12m | Pre-decision | Credit-seeking behavior |
| open_acc | Pre-decision | Credit profile |
| total_acc | Pre-decision | Credit history |
| revol_bal | Pre-decision | Credit exposure |
| revol_util | Pre-decision | Credit utilization |
| installment | Pre-decision | Repayment burden |
| mort_acc | Pre-decision | Credit profile |
| pub_rec | Pre-decision | Credit risk |
| pub_rec_bankruptcies | Pre-decision | Credit risk |
| collections_12_mths_ex_med | Pre-decision | Credit risk |
| num_accts_ever_120_pd | Pre-decision | Historical severe delinquency |
| pct_tl_nvr_dlq | Pre-decision | Credit behavior |
| num_sats | Pre-decision | Credit profile |
| tot_coll_amt | Pre-decision | Historical credit risk |
| tot_cur_bal | Pre-decision | Current credit exposure |
| tot_hi_cred_lim | Pre-decision | Credit capacity |
| total_bal_ex_mort | Pre-decision | Debt exposure |
| total_bc_limit | Pre-decision | Revolving credit capacity |
| bc_util | Pre-decision | Bankcard utilization |
| all_util | Pre-decision | Overall credit utilization |
| loan_amnt | Decision / Policy | Lending-policy arm |
| int_rate | Decision / Policy | Lending-policy arm |
| term | Decision / Policy | Lending-policy characteristic |
| grade | Pre-decision | Credit-risk context |
| sub_grade | Pre-decision | Credit-risk context |
| funded_amnt | Decision / Disbursement | Actual loan exposure |

| Variable | Availability Stage | Primary Use |
|---|---|---|
| issue_d | Decision / Disbursement | Loan origination and time index |
| last_fico_range_low | Interim | Updated borrower risk |
| last_fico_range_high | Interim | Updated borrower risk |
| last_pymnt_amnt | Interim | Interim repayment reward |
| last_pymnt_d | Interim | Repayment timing |
| out_prncp | Interim | Remaining exposure |
| out_prncp_inv | Interim | Remaining investor exposure |
| num_tl_30dpd | Interim | Current delinquency |
| num_tl_90g_dpd_24m | Interim | Evolving credit risk |
| mths_since_last_delinq | Interim | Evolving delinquency risk |
| next_pymnt_d | Interim | Scheduled repayment information |
| loan_status | Terminal | Default and credit-state outcome |
| recoveries | Terminal | Recovery estimation |
| collection_recovery_fee | Terminal | Recovery/revenue calculation |
| total_pymnt | Terminal | Total repayment |
| total_pymnt_inv | Terminal | Investor repayment |
| total_rec_prncp | Terminal | Principal recovered |
| total_rec_int | Terminal | Interest revenue |
| total_rec_late_fee | Terminal | Late-fee revenue |

### *F.2 Data Availability by Modeling Stage*

The retained variables will be used differently across the proposed framework. **Pre-decision variables** will define the borrower context available when a lending policy is considered. The loan amount, interest rate, and term will define the candidate lending policy. **Interim variables** will become available as the loan progresses and will be used to update borrower-state estimates and delayed rewards. **Terminal variables** will only be used once the corresponding loan outcomes become available.

*Table 4: Data Availability Across the Proposed Framework*

| Modeling Stage | Information Available | Framework Component |
|---|---|---|
| **Pre-decision** | Borrower characteristics, credit history, income, FICO, DTI, employment, loan purpose | Contextual risk estimation |
| **Policy selection** | Pre-decision context + candidate loan_amnt, int_rate, and term | Contextual Thompson Sampling |
| **Interim** | Repayment, delinquency, outstanding principal and updated credit information | MCTM and interim reward estimation |
| **Terminal** | loan_status, recoveries, total payments and realized revenue | Default/recovery modeling and final reward |
| **Policy evaluation** | Observed outcomes and posterior predictive estimates | Shadow lending engine and policy comparison |

### *F.3 Prevention of Data Leakage*

The temporal classification of variables will be used to prevent future information from influencing earlier lending decisions. In particular, terminal variables such as loan_status, recoveries, and total_pymnt will not be provided to the contextual Thompson Sampling model when it selects a lending policy.

Instead, these variables will become observations that are used to update the Bayesian risk and reward distributions after the relevant outcomes have occurred. Similarly, interim variables will only be introduced into the model at the point in the loan life cycle at which they would have become available.

### *F.4 Data Privacy and Minimization*

The study will apply data minimization by excluding variables that are not necessary for the proposed analysis, particularly direct identifiers, free-text fields, URLs, and unnecessarily granular geographic information. The retained variables are selected primarily for their relevance to borrower risk, lending-policy selection, credit-state transitions, repayment behavior, and policy evaluation.

The retained financial and credit information will be treated as research data and will not be used to identify, contact, or make actual lending decisions concerning individual borrowers. The resulting

framework is intended as a research prototype for evaluating Bayesian sequential decision-making in credit risk modeling.

# Appendix G: Experimental Protocols

## *G.1 Bayesian Risk Modeling Protocol*

The first experiment will evaluate the ability of the hierarchical Bayesian framework to estimate borrower default probability and conditional recovery. The model will use the pre-decision borrower characteristics identified in Appendix F and will separate default estimation from recovery estimation.

The first stage will estimate the probability of default using a hierarchical Bayesian Beta-Bernoulli formulation. The second stage will estimate recovery conditional on default using a Beta-based model. Posterior predictive distributions will subsequently provide uncertainty estimates for borrower risk and expected recovery.

*Table 5: Proposed Bayesian Risk Modeling Protocol*

| Component | Proposed Procedure |
|---|---|
| Input | Pre-decision borrower and loan variables |
| Stage 1 | Estimate probability of default |
| Stage 2 | Estimate recovery conditional on default |
| Hierarchical structure | Share information across borrower groups |
| Output | Posterior distributions for default and recovery |
| Validation | Brier score, ROC-AUC, calibration and recovery error |
| Uncertainty | Posterior credible intervals and posterior predictive distributions |

## *G.2 Markov Chain Transition Matrix Protocol*

The second experiment will evaluate the ability of the **Markov Chain Transition Matrix (MCTM)** to model borrower behavior throughout the loan life cycle.

Borrowers will be represented using four credit states: **Current, Delinquent, Recovered, and Absorbed Loss**. Current and Delinquent will represent transitional states, while Recovered and Absorbed Loss will be treated as absorbing states.

The transition probabilities will be estimated from the available repayment and delinquency information and evaluated according to their ability to represent observed borrower-state transitions.

*Table 6: Proposed MCTM Credit States and Definitions*

| State | Description | Type |
|---|---|---|
| **Current** | Borrower is meeting the expected repayment obligations | Transient |
| **Delinquent** | Borrower has fallen behind on repayment obligations | Transient |
| **Recovered** | Loan has resulted in repayment/recovery following delinquency | Absorbing |
| **Absorbed Loss** | Loan has resulted in a final loss/default outcome | Absorbing |

The MCTM will provide time-dependent estimates of the probability that a borrower moves between these states. These estimates will subsequently contribute to the compound risk-adjusted reward used by the policy-selection component.

### *G.3 Contextual Thompson Sampling Protocol*

The third experiment will evaluate contextual Thompson Sampling for sequential lending-policy selection. Each arm will represent a candidate lending policy defined primarily by a combination of **loan amount and interest rate**, with loan term incorporated where appropriate.

At each decision point, the algorithm will receive the available borrower context and sample from the posterior reward distribution associated with each candidate policy. The policy with the highest sampled reward will be selected.

*Table 7: Proposed Contextual Bandit Experimental Configuration*

| Component | Proposed Configuration |
|---|---|
| Context | Pre-decision borrower characteristics |
| Arms | Candidate combinations of loan amount, interest rate and term |
| Policy | Contextual Thompson Sampling |
| Reward | Compound risk-adjusted financial reward |

| Component | Proposed Configuration |
|---|---|
| Exploration | Posterior sampling |
| Exploitation | Selection of policies with high posterior expected reward |
| Feedback | Interim and terminal repayment outcomes |
| Updating | Bayesian posterior update after observed feedback |
| Evaluation | Cumulative reward and cumulative regret |

The algorithm will be evaluated against appropriate baseline strategies, including the historically observed lending policy where feasible. The objective will be to determine whether adaptive policy selection can improve risk-adjusted outcomes rather than simply maximizing loan approvals or minimizing defaults.

### *G.4 Counterfactual Shadow Lending Policy Evaluation*

Because historical data only records outcomes for lending policies that were actually applied, the study will incorporate a **shadow lending engine** to estimate the potential outcomes of alternative policies.

For a given borrower context, the engine will generate posterior predictive estimates under alternative loan amounts and interest rates. These estimates will be used to construct counterfactual risk-adjusted rewards without representing the simulated policy as an observed historical outcome.

*Table 8: Proposed Counterfactual Policy Evaluation*

| Component | Procedure |
|---|---|
| Historical borrower context | Used as the baseline |
| Historical policy | Used as the observed reference |
| Alternative policy | Candidate loan amount, interest rate and term |
| Default | Estimated from Bayesian posterior |
| Recovery | Estimated conditional on default |
| Credit transitions | Estimated using MCTM |
| Financial reward | Calculated from predicted repayment and exposure |
| Policy comparison | Historical versus counterfactual policies |

| Component | Procedure |
|---|---|
| | |
| Uncertainty | Posterior predictive intervals |

The shadow lending engine will therefore provide a controlled method for comparing policies that were not necessarily observed in the historical dataset.

## *G.5 Distributional Shift Monitoring Protocol*

A Population Stability Index (PSI) Sentinel will be incorporated to monitor whether the borrower population encountered during sequential decision-making differs materially from the population used during model development.

PSI will be calculated for selected contextual variables and monitored over time. Proposed thresholds will be treated as experimental monitoring criteria rather than universal definitions.

*Table 9: Distributional Shift Monitoring Protocol*

| PSI Range | Proposed Interpretation | Planned Response |
|---|---|---|
| < 0.10 | Little or no distributional shift | Continue normal operation |
| 0.10–0.25 | Moderate shift | Increase monitoring and review |
| > 0.25 | Significant shift | Trigger Bayesian review/circuit breaker |

Where substantial distributional changes are detected, the framework will require additional model assessment before continued policy optimization. Human review will be incorporated into this process rather than allowing the monitoring mechanism to independently authorize changes to real lending decisions.

## *G.6 Infrastructure Resilience Protocol*

The proposed prototype will be evaluated under the resource constraints of the target bare-metal Kubernetes environment. Experiments will examine whether the distributed system can continue operating or recover appropriately when individual computational resources become unavailable.

*Table 10: Proposed Infrastructure Resilience Experiments*

| Experiment | Failure Scenario | Evaluation |
|---|---|---|
| Node failure | One worker node becomes unavailable | Workload recovery |
| Multiple-node failure | Multiple workers become unavailable | System resilience |
| Workload termination | Running computation is canceled | Workflow recovery |
| Resource constraint | CPU or memory becomes constrained | Computational stability |
| Network interruption | Temporary network connectivity loss | Service recovery |
| Controlled power interruption | Safe simulation of node power loss | Recovery behavior |
| State recovery | Computation interrupted during processing | Kafka/event-state recovery |

Where possible, failure scenarios will be simulated through controlled software or system operations rather than unsafe physical interventions.

### *G.7 Overall Evaluation Protocol*

The final experiment will evaluate the framework across four dimensions: **predictive performance, decision-making performance, adaptability, and computational feasibility**.

*Table 11: Proposed Evaluation Metrics*

| Evaluation Dimension | Metrics |
|---|---|
| **Default prediction** | Brier Score, ROC-AUC, calibration error |
| **Recovery prediction** | MAE, RMSE |
| **Credit-state modeling** | Transition likelihood and transition prediction error |
| **Policy performance** | Risk-adjusted profitability, cumulative reward |
| **Bandit performance** | Cumulative regret, reward convergence, policy selection frequency |
| **Counterfactual evaluation** | Historical versus simulated policy performance |
| **Distributional monitoring** | PSI and frequency of detected shifts |
| **Computational feasibility** | CPU utilization, memory utilization, execution time |
| **Infrastructure resilience** | Recovery time, failed workloads, successful state recovery |

The proposed experiments will use appropriate training, validation, and testing procedures to avoid evaluating the framework on information that was available during model development. Results will be compared against relevant baseline approaches where applicable.

Overall, the experimental protocol is designed to determine whether the proposed combination of hierarchical Bayesian risk estimation, MCTM, contextual Thompson Sampling, counterfactual simulation, and distributional monitoring can provide an adaptive and computationally feasible approach to sequential credit-policy optimization.